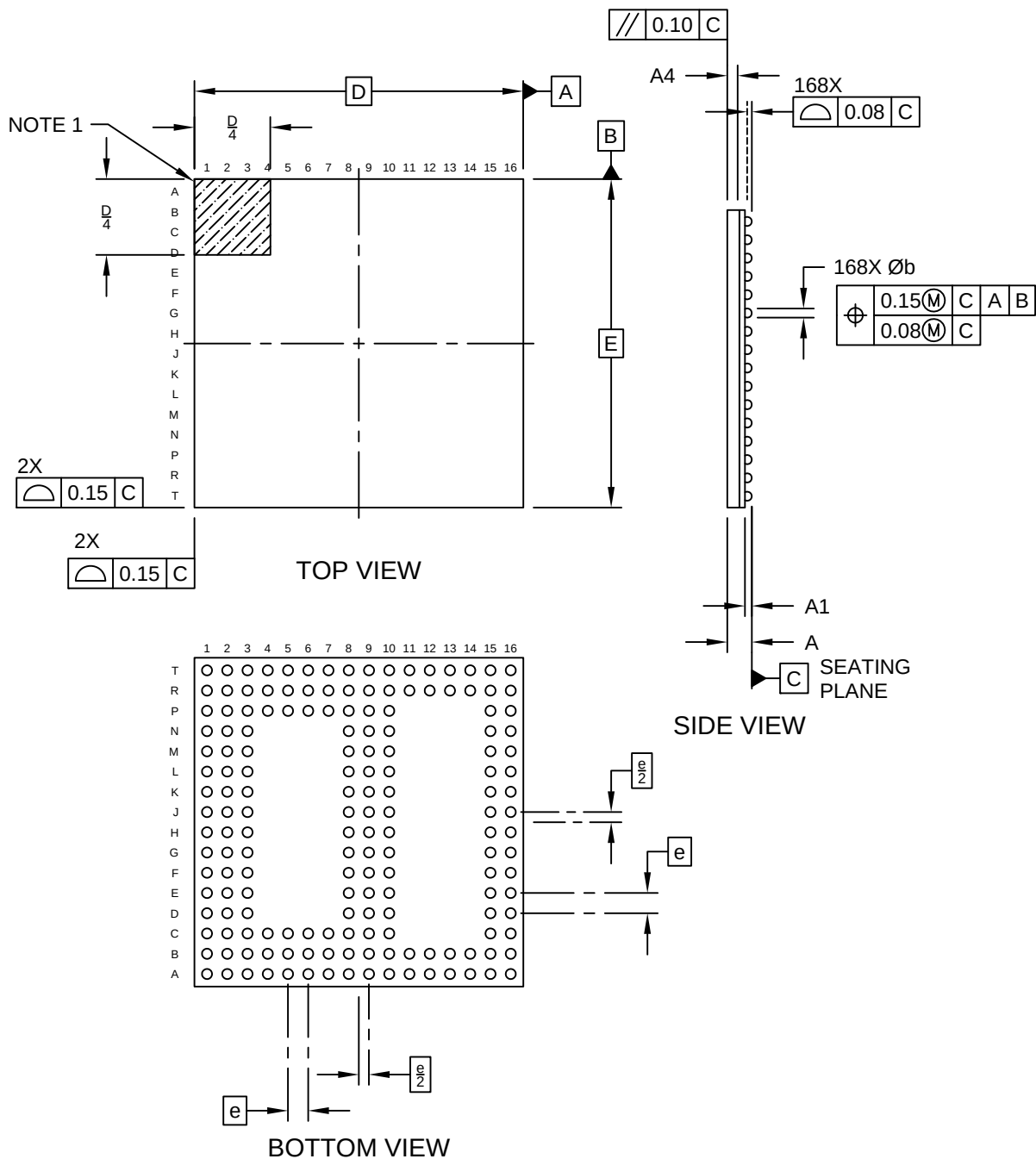


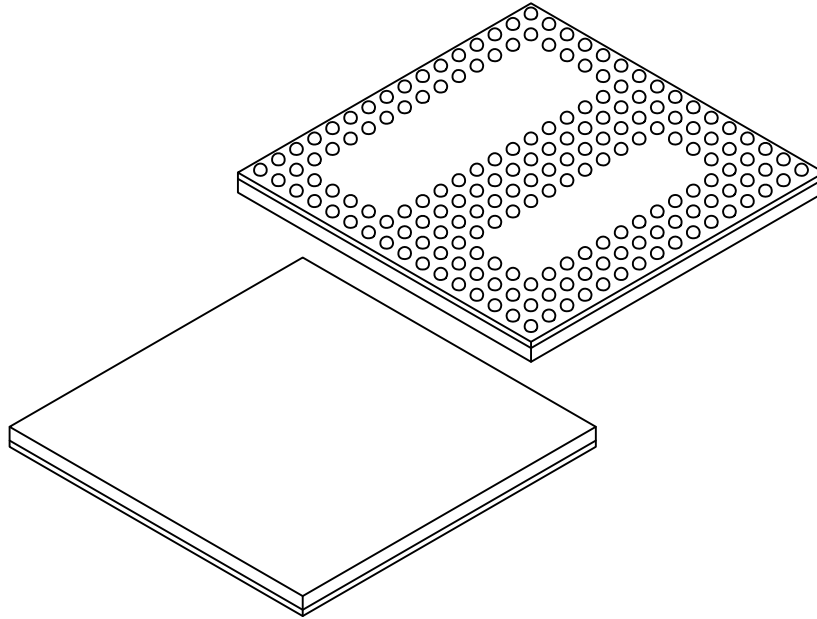
168-Ball Thin Fine-Pitch Ball Grid Array (AFA) - 13x13x1.2 mm Body [TFBGA]
Internal Flip chip

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



168-Ball Thin Fine-Pitch Ball Grid Array (AFA) - 13x13x1.2 mm Body [TFBGA] Internal Flip chip

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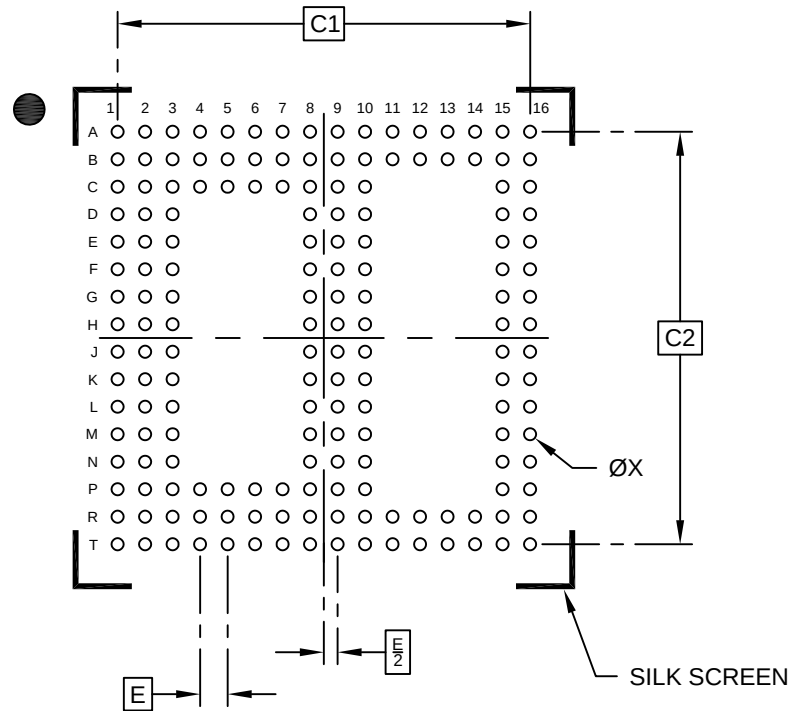
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	168		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.20
Standoff	A1	0.23	0.33	-
Mold Cap Height	A4	0.53 REF		
Overall Length	D	13.00 BSC		
Overall Width	E	13.00 BSC		
Ball Diameter	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

168-Ball Thin Fine-Pitch Ball Grid Array (AFA) - 13x13x1.2 mm Body [TFBGA] Internal Flip chip

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Overall Contact Pad Spacing	C1	12.00 BSC		
Overall Contact Pad Spacing	C2	12.00 BSC		
Pad Diameter (X168)	ØX			0.35

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.